

# Lab 06

## spice

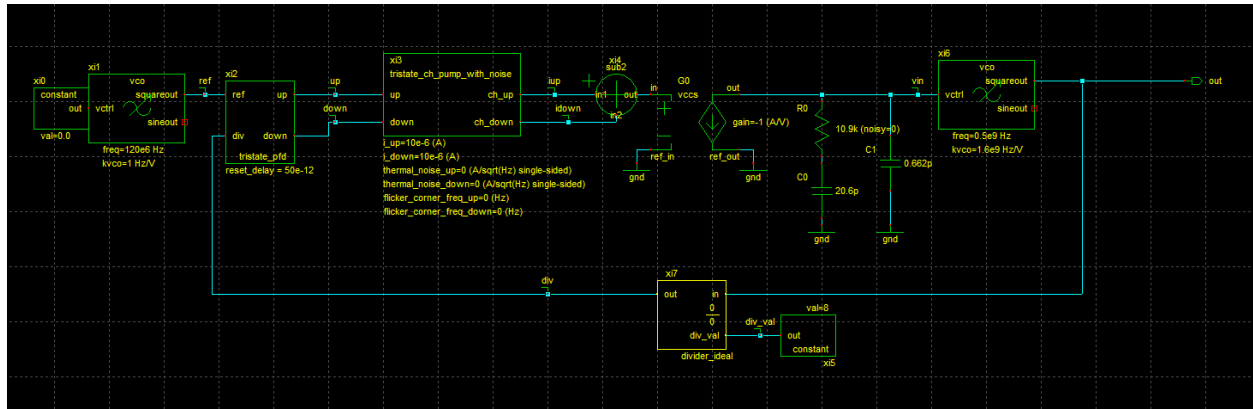
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# PART1 (PRELAB)

## 1 SCHEMATIC OF THE PLL DRAWN IN SUE2.



## 2 SNAPSHOT OF THE RANDOM NUMBER GENERATION WEBSITE SHOWING YOUR RANDOM NUMBER. CALCULATE KVCO.

160

Lower Limit

Upper Limit

Generate

$$K_{VCO} = \frac{N}{100} * 10^9 = 1.6 \text{ GHz/V}$$

### 3 GIVEN $K_{VCO}$ , ANALYTICALLY CALCULATE $V_{IN}$ AT WHICH THE PLL WILL ACHIEVE LOCK.

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We have frequency divider modules equals 8, and the input frequency equals 120GHZ, so the output frequency equals 960GHZ, though as we see from the VCO model, it has a constant initial frequency equals 500GHZ adds to the VCO function itself, so  $960-500=460\text{MHZ}$ , this is the output frequency from the pure VCO:

$$K_{VCO} = \frac{F}{V_{in}}, \text{ so } V_{in} = \frac{460}{1.6} = 287.5 \text{ mV}$$

### 4 SNAPSHOT OF CPPSIM RUN DIALOG SHOWING SUCCESSFUL SIMULATION COMPLETION.

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```
ult: ..... 40% completed .....
      ..... 50% completed .....
      ..... 60% completed .....
      ..... 70% completed .....
      ..... 80% completed .....
      ..... 90% completed .....

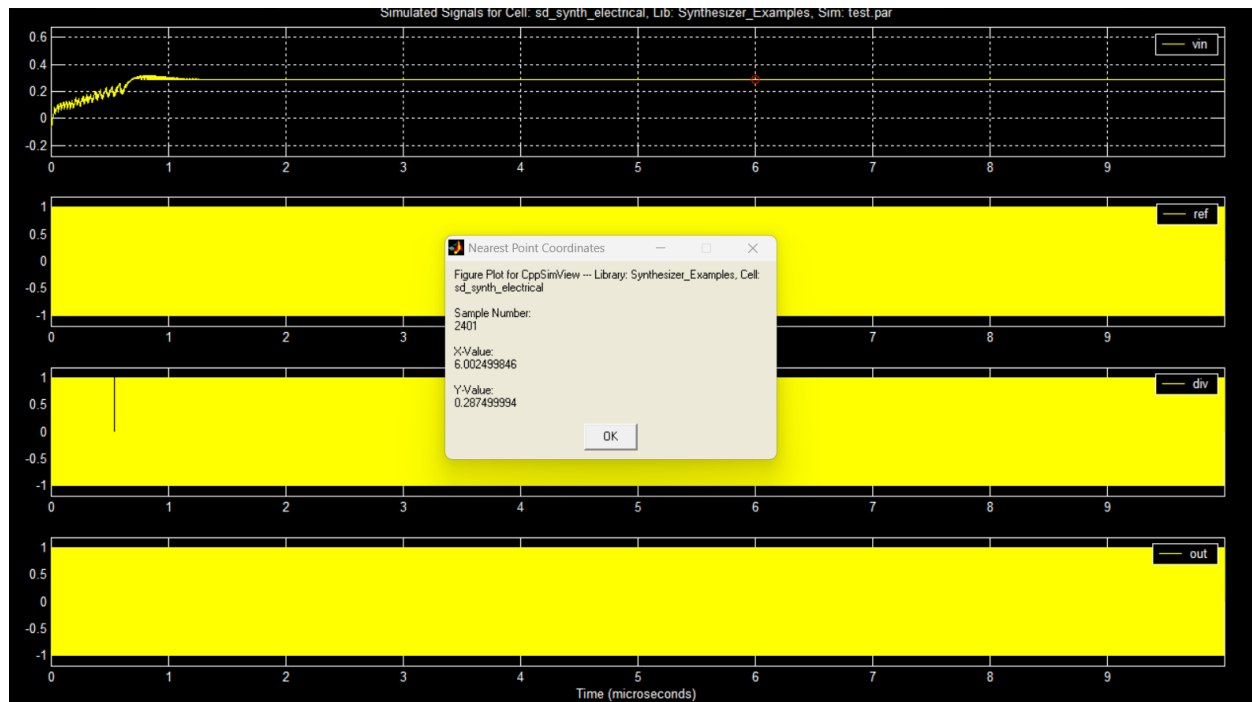
Simulation Completed for Alter Run 0

***** CppSim run completed! *****

Note: run files contained in directory:
      C:/CppSim/SimRuns/Synthesizer_Examples/sd_synth_electrica
```

## 5 SNAPSHOT OF THE SIMULATION RESULTS WITH A PROBE SHOWING VIN VALUE AS SHOWN ABOVE

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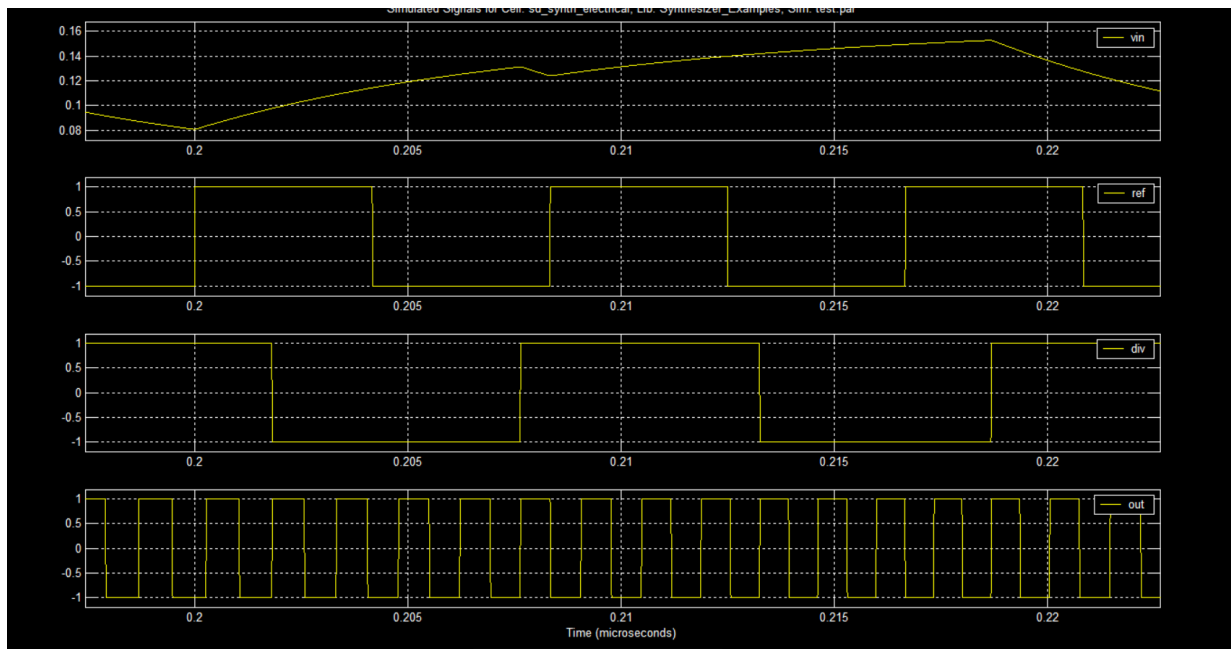


It locked at  $V_{in} = 287.5 \text{ mV}$

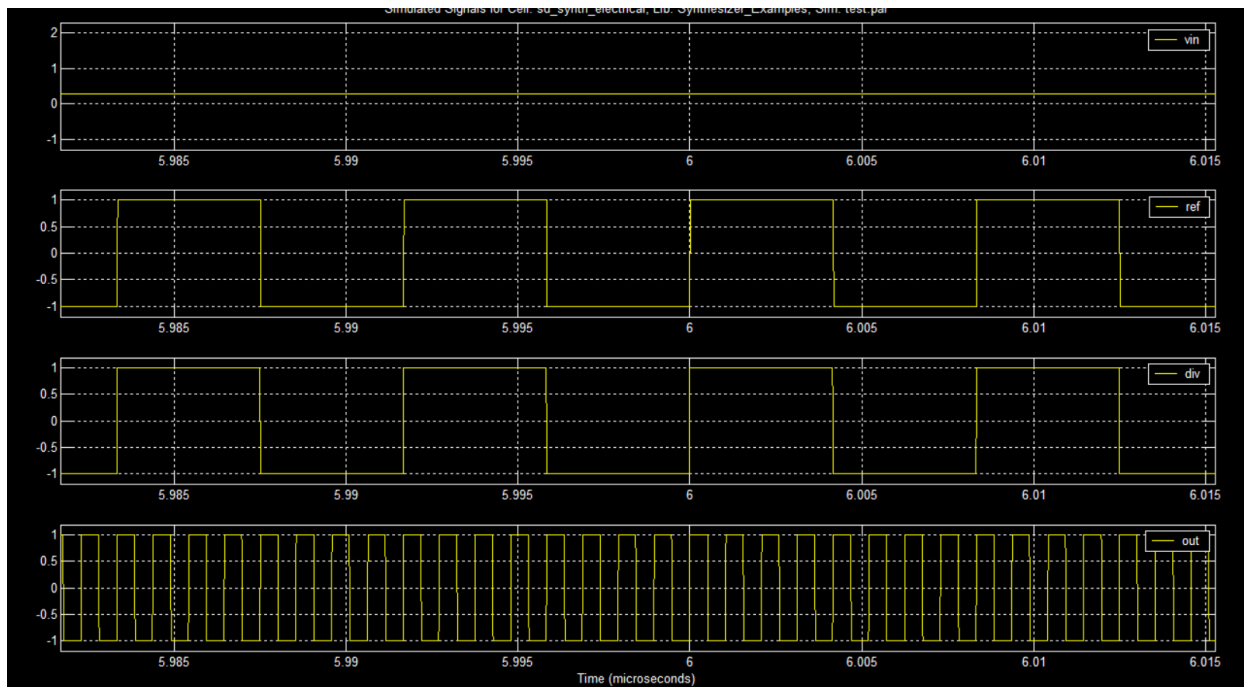
## 6 COMPARE THE SIMULATED VIN VALUE WITH THE ANALYTICALLY CALCULATED VALUE. COMMENT.

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There are the same equals 287.5mv



## 7 CLEAR ZOOM X SNAPSHOT AFTER LOCK AS SHOWN ABOVE.

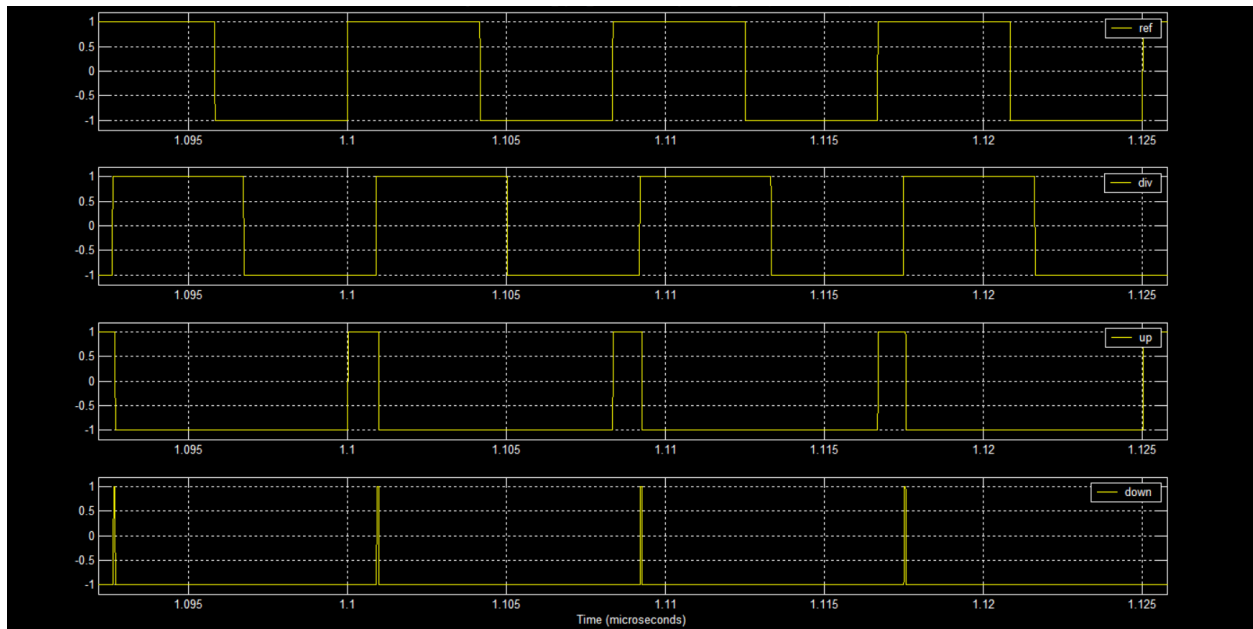


## PART2

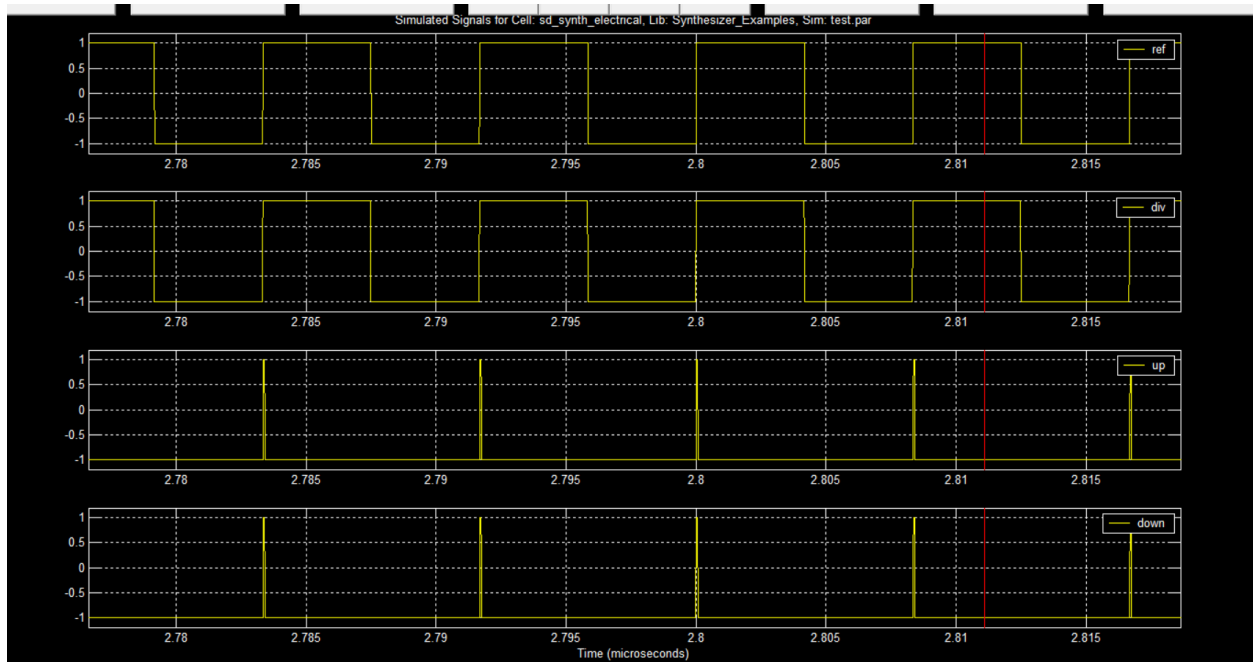
### 1 PLOT (REF,DIV,UP,DOWN). REPORT CLEAR ZOOM X SNAPSHOT SHOWING THE PFD OPERATION.

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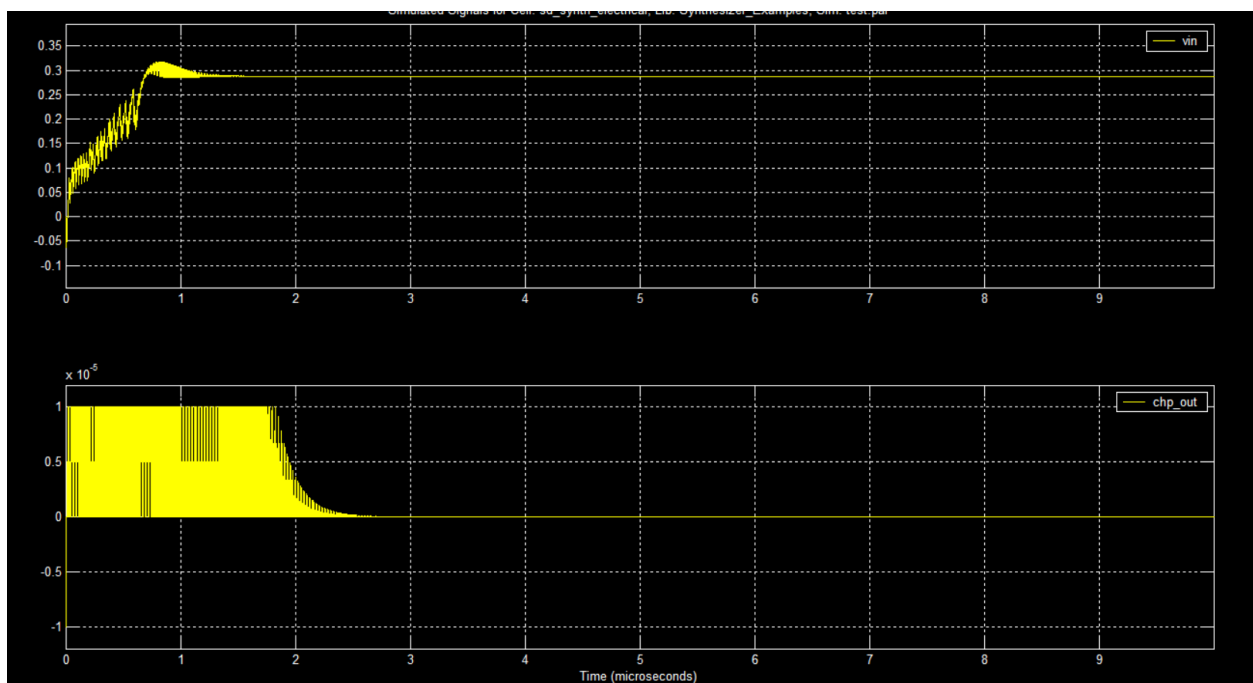
Ref came before div so 'up' did it's operation, and produced a pulse its width is the difference between the ref and div signals.

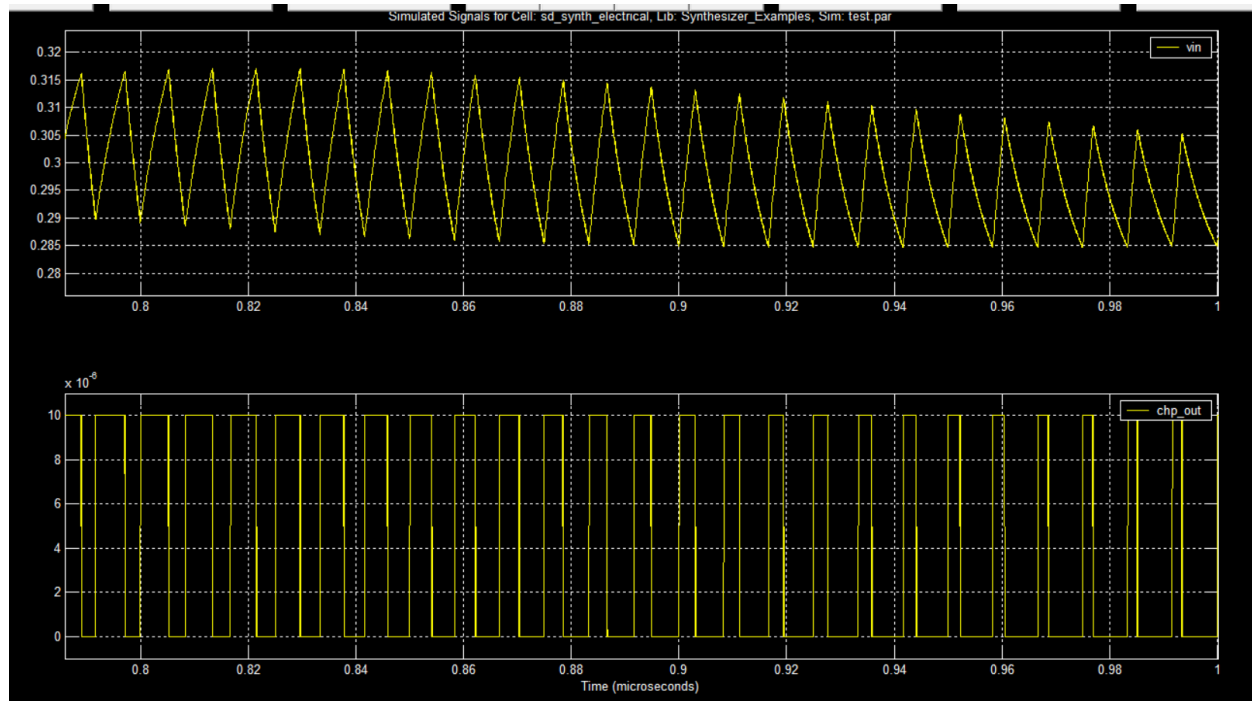


When 'ref' and 'div' come together (locked phases), we have 'up' and 'down' pulses with negligible widths.



## 2 RESET NODE LIST. PLOT (VIN,CHP\_OUT).

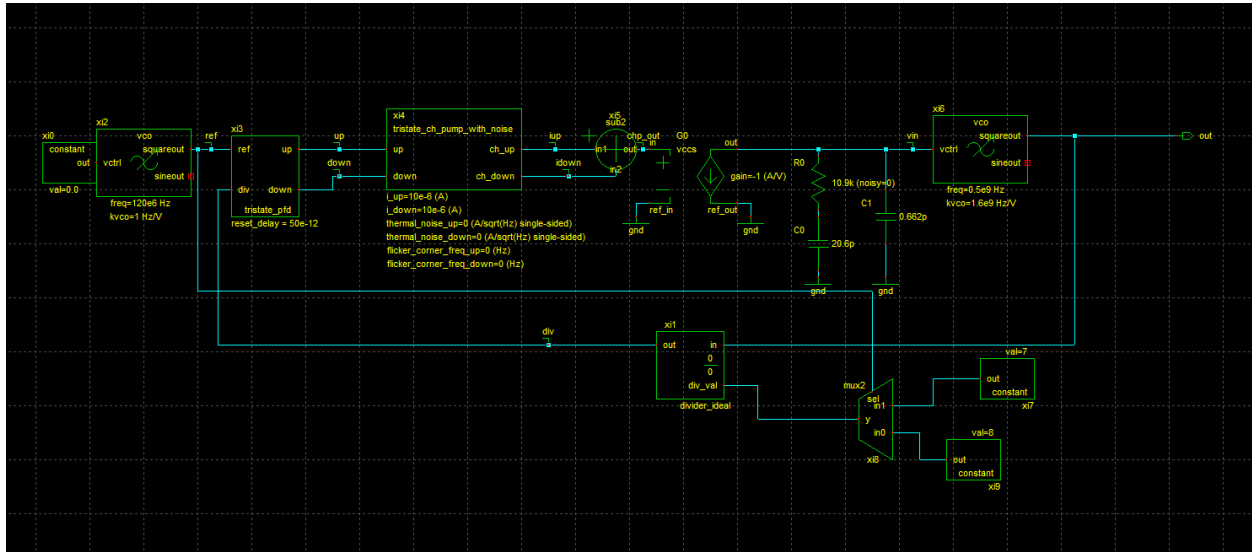




We had high pulses before locking, but after locking we have negligible pulses as we have locked phases ( $\Phi_{in} = \Phi_{out}$ ).

## PART3

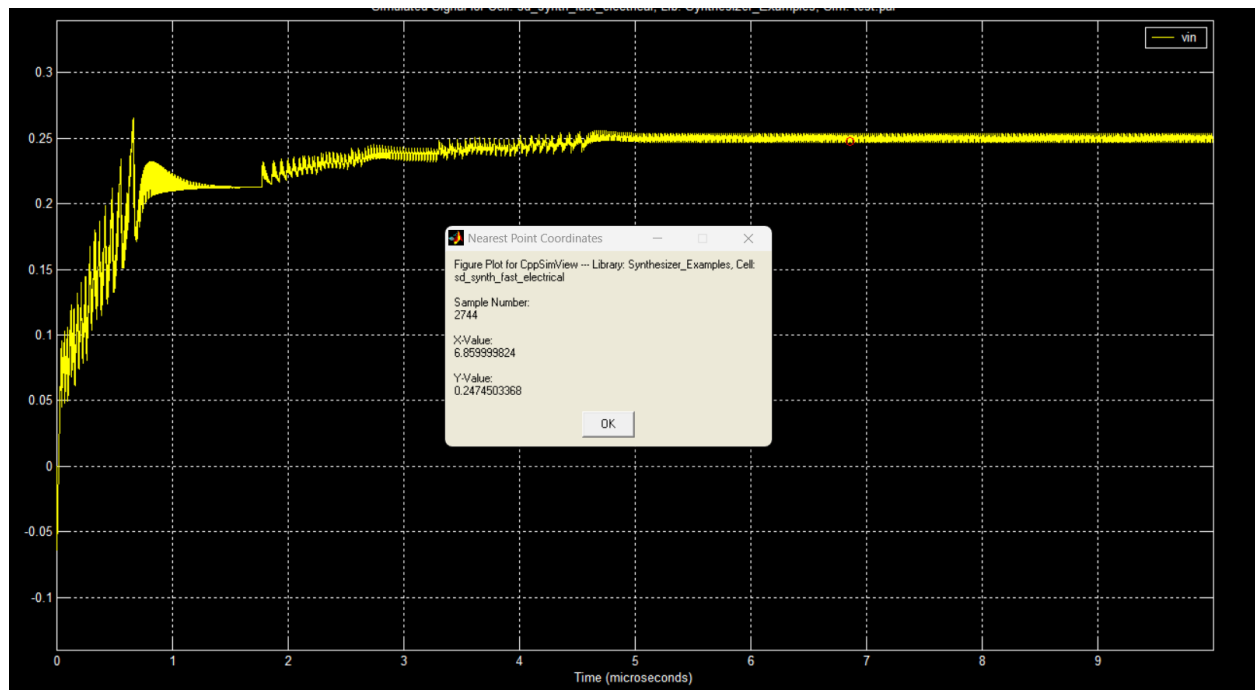
- 1 GIVEN KVCO AND THE NEW DIVIDER VALUE, ANALYTICALLY CALCULATE VIN AT WHICH THE PLL WILL ACHIEVE LOCK.



$$f_{out} = 120 * \frac{8+7}{2} = 900MHz$$

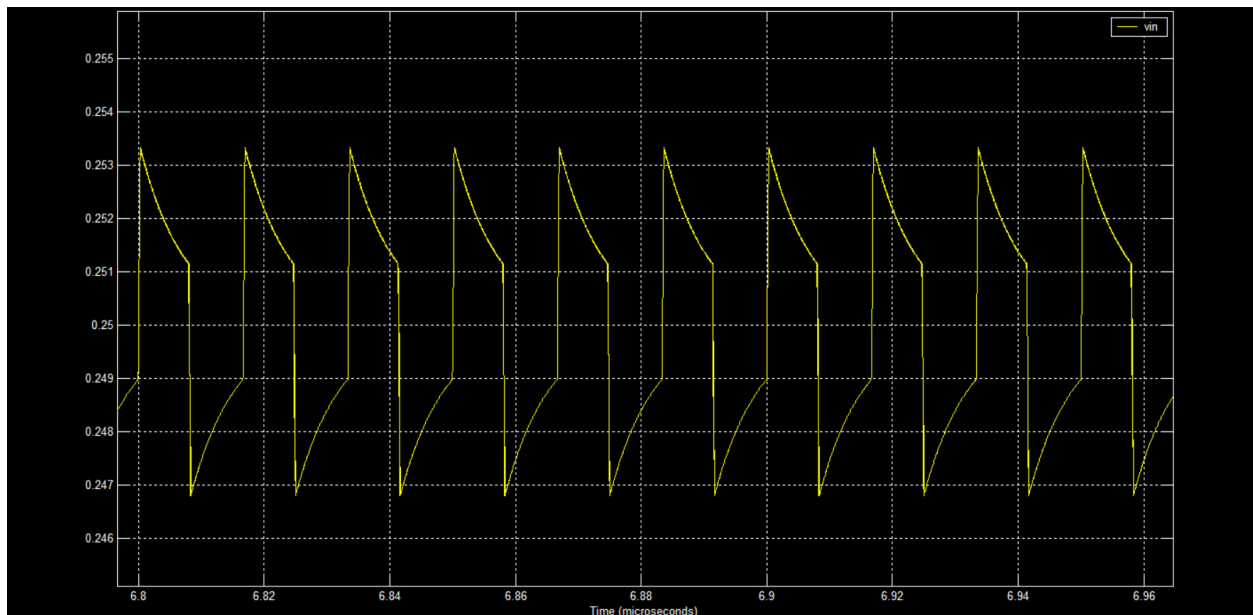
$$K_{VCO} = \frac{f}{V_{in}}, so V_{in} = \frac{900 - 500}{1.6} = 250 mV$$

## 2 SNAPSHOT OF THE SIMULATION RESULTS SHOWING VIN WAVEFORM AFTER LOCKING. COMMENT.



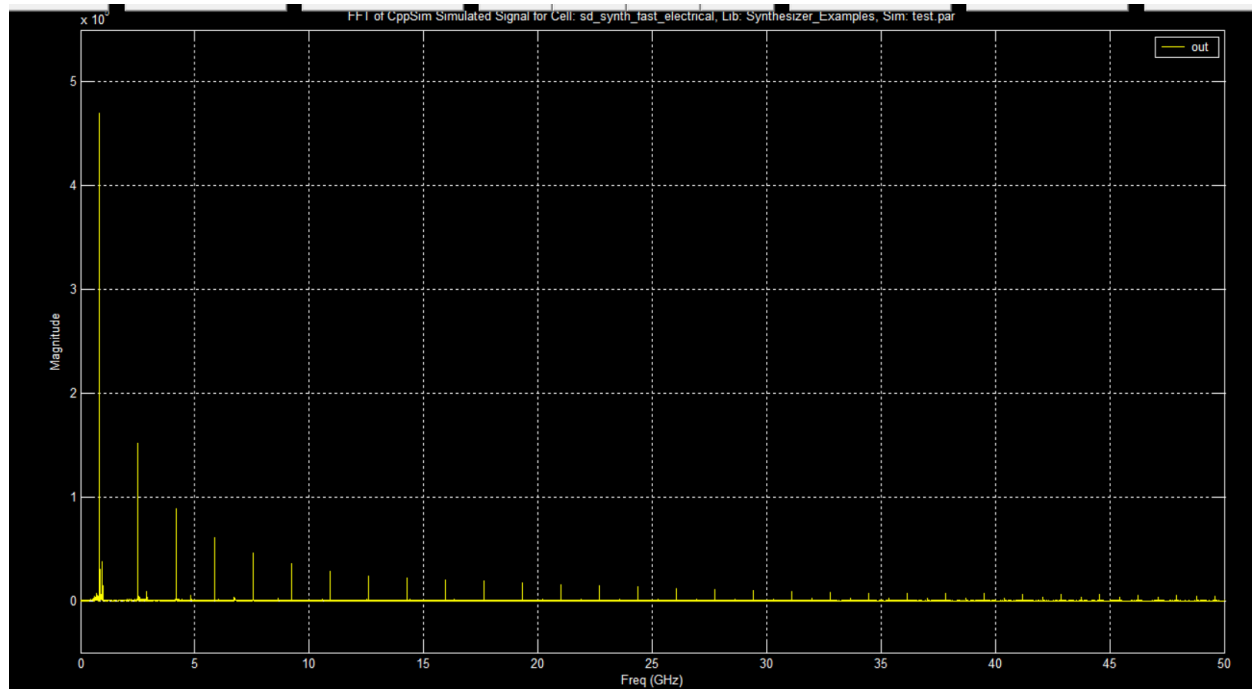
$$V_{in} = 247.45 \text{ mV}$$

## 2.1 COMMENT:



We actually don't have absolute constant  $v_{in}$  as it alternates between 247mv to 253mv, this is also expected as we alternate between  $M=8$  and  $M=9$  and it depends on the speed of loop, if the loop is faster we would see less variations.

### 3 USE CPPSIMVIEW PLOT FUNCTIONS TO PLOT THE OUTPUT SPECTRUM (FFT). COMMENT. FOR HELP, IN SUE2 CLICK (DOC -> CPPSIM/VPPSIM PRIMER).



#### 3.1 COMMENT

the main component (with the highest magnitude) is at 900MHZ, but as shown we have other components beside the fundamental as it is a square wave not sinusoidal wave, also we have other components as VIN is not constant so we have other frequencies.